A Bounded First Rung toward "Electrical fMRI": Subject-Transferable Decoding and Intracranial Coupling of a Medial-Temporal-Lobe Working-Memory State from Scalp EEG

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Abstract

The long-term goal that motivates this work is "electrical fMRI": recovering a spatially- and temporally-resolved picture of deep-brain activity from cheap, accessible scalp EEG plus machine learning. That goal spans many projects. This Station scoped and executed the smallest sufficient *first* claim that makes verifiable progress toward it, using the Boran et al. (2019) open dataset of *simultaneous* scalp EEG, intracranial EEG/LFP, and medial-temporal-lobe (MTL) single-unit recordings from nine epilepsy patients performing a verbal working-memory (Sternberg) task. We asked whether scalp EEG carries a *subject-transferable*, intracranially-validated signature of MTL working-memory *load*, evaluated leave-one-subject-out (LOSO). We report a deliberately two-part, pre-registered result. **Part A (a clean negative).** Across a pre-registered ladder of model classes—regularized logistic regression, filter-bank covariance, Riemannian (tangent-space and minimum-distance-to-mean) classifiers, and a compact EEGNet-class CNN—an 8-channel common montage does not yield a load decoder that beats the strongest non-signal control by the pre-declared margin (+0.075 balanced accuracy) under LOSO. The best model (EEGNet) reaches +0.023, with the entire positive mean carried by a single subject, failing the pre-declared robustness clause. The binding constraint is the 8-channel common montage plus cross-subject transfer, not the model class. **Part B (an exploratory lead, not validated).** The intracranial MTL theta-minus-alpha load substrate is real (7/9 subjects, two-sided $p = 0.016$), and the strongest scalp decoder's output shows a *raw* positive coupling to it (7/9, $p \approx 0.05$) that the weaker linear decoders do not—but that coupling does not survive load/schedule residualization, so at $n = 9$ it cannot be cleanly distinguished from a shared load- or task-schedule-linked correlate. Both outcomes were named as pre-declared success/failure shapes before any result was observed. The result tells the larger electrical-fMRI program where the first wall is and names the most plausible next signal to test with more statistical power.

1 Introduction

When neurons fire, they generate electric fields that propagate through brain tissue, skull, and scalp. Scalp EEG measures a blurred, summed projection of that activity from outside the head. Deep structures such as the medial temporal lobe (MTL), central to memory, are conventionally treated as nearly invisible to scalp EEG: the skull is roughly two orders of magnitude less conductive than brain tissue, deep fields are faint and geometrically ambiguous at the surface, and the EEG inverse problem is ill-posed for deep sources. Yet deep structures *couple* to cortex during cognition. In verbal working memory specifically, hippocampal firing and hippocampal–cortical theta–alpha synchronization scale with memory load [2], and that coupling reaches the scalp: hippocampal stereo-EEG and scalp EEG show measurable theta phase-locking during maintenance [6]. The bet behind this project is that such coupling leaves an indirect but *learnable* signature in scalp EEG, even where the deep fields themselves are not directly measurable at the surface.

The north star is "electrical fMRI": using cheap scalp EEG plus AI to reconstruct a coarse, spatially-resolved picture of deep activity that today requires expensive, immobile fMRI. That is a multi-project arc. The job of this Station was to choose and execute the *first* rung—the smallest standalone claim that makes real, checkable progress—rather than to attempt the whole staircase. What makes the first rung checkable is *validation data*: the Boran et al. (2019) dataset [3] provides *simultaneous* scalp EEG, intracranial EEG/LFP, and MTL single-unit recordings from the same brain at the same moment. We can take scalp EEG as input and check inferences about deep activity against directly recorded intracranial truth. That simultaneity is what turns "infer deep activity from the scalp" from speculation into a held-out-testable measurement.

The first-rung question. Does scalp EEG carry a *subject-transferable* signature of an intracranially-validated MTL working-memory state—specifically working-memory *load*—that a model can read out from the scalp alone, on a person it has never seen, above strong non-neural controls, and that is mechanistically tied to the MTL theta–alpha coupling the intracranial data validates? Load is the target because it is a task variable defined independently of any neural channel, available for all nine subjects, and easy for a non-specialist to audit.

Contributions. This report contributes a rigorously characterized, two-part answer, each part pre-registered as an allowed outcome before any result was seen (Section 3.3):

1. **A clean negative boundary (Part A).** Across a complete, pre-registered model ladder, an 8-channel common scalp montage does not support subject-transferable load decoding above the strongest non-signal control by the pre-declared margin. We localize the binding constraint to the common montage plus cross-subject transfer, not to model capacity.
2. **An exploratory mechanism lead (Part B).** The intracranial MTL theta-minus-alpha load substrate is real, and the strongest scalp decoder's output couples to it in the raw trial summaries where weaker decoders do not—but the coupling does not survive stricter residualization. We report this as exploratory, not as a validated deep readout, and name it as the most plausible next signal to test with more power.
3. **Reusable, efficiency-constrained infrastructure** (NIX reader, epoch alignment, LOSO harness, a hand-rolled feature/Riemannian/EEGNet library that runs on a consumer laptop without a deep-learning framework, a control/statistics harness, and a per-subject verification dashboard) that every later rung of the program reuses.

A clean negative is a deliberate, valuable artifact here: it tells the larger program that this dataset and montage cannot support a transferable first claim as scoped, redirecting the staircase early rather than after a much larger investment.

2 Dataset and substrate

We use the Boran et al. (2019) open dataset [3, 4], distributed under CC BY-SA 4.0 via G-Node (DOI `10.12751/g-node.d76994`) in the NIX/HDF5 format (37 session files). The raw data is never committed to this project's repository; the Reproducibility Packet references the public DOI and instructs the reader to download it. The dataset comprises, for nine epilepsy patients performing a modified Sternberg verbal working-memory task with temporally separated encoding, maintenance, and recall periods:

- **Scalp EEG** (10–20 montage)—the cheap, accessible signal that is this project's input.
- **Intracranial EEG / LFP** (depth electrodes in and around the MTL)—the deep "ground truth" not normally available, used here only as a validation channel, never as a decoder input.
- **Single-unit activity** (1526 MTL neurons; spike times and waveforms).
- **MNI coordinates and anatomical labels** for all intracranial contacts, plus per-trial behavioral metadata (set size, match/mismatch, correct/incorrect, reaction time).

The defining property is *simultaneity*: scalp and deep signals from the same brain at the same moment. The task structure also carries a rule that becomes important in Section 4.1: an incorrect response is followed by a forced low-set-size (set size 4) trial, which couples the previous trial's correctness to the current trial's load.

Headline target and epoch. The headline analysis decodes binary working-memory load — set size 4 (low) versus set sizes 6/8 (high), matching the Boran-style low/high contrast—from the **maintenance period**. Maintenance is chosen because it isolates a *maintained* working-memory state; an encoding-period model could win by reading transient sensory stimulus-load cues rather than the maintained deep state of interest. All window and channel choices were fixed inside the training subjects only.

Common montage. To make a single model transfer across subjects, decoding uses an 8-channel common montage shared by all nine subjects (the headline "all" channel set), with a 6-channel "brain-only" subset (excluding the A1/A2 ear references) reserved as a pre-declared reference-artifact sensitivity check.

3 Methods

All software follows the project's engineering standards: every script has a single purpose, takes machine-specific paths via `argparse` (`required=True`, no hard-coded paths), shares logic through a `utils/` module, prints progress, writes named outputs, and fails loudly on bad input. Under an efficiency constraint (an 8 GB-VRAM consumer laptop, 16 GB RAM), the Riemannian geometry and the EEGNet CNN are implemented dependency-free in NumPy rather than pulling in heavyweight frameworks; both are validated against analytic checks before use (Section 3.2).

3.1 Data layer and epoching

A NIX reader (`utils/nix_io.py`, built on `nixio` [7]) loads scalp epochs and, lazily, the intracranial epochs used only by the mechanism layer. The reader was validated against the dataset's documented structure as a stop-or-go correctness gate. One data quirk drove a design choice: the files use NIX's legacy metadata format, and the reader decodes character properties with a per-property safe-read so that a single non-UTF-8 metadata field (e.g. a German place-name) cannot abort a whole-session read. Maintenance-window epochs are extracted on the common montage; adjacent windows from the same trial/subject never straddle the train/test boundary (an autocorrelation-leakage guard).

3.2 Features and the pre-registered model ladder

The model ladder is ordered smallest-sufficient-first; each rung is a distinct, pre-registered model class, and all share the same LOSO folds, the same retained trials, and the same output contract so that the control and statistics harness consumes every rung identically.

1. **Regularized logistic / linear** on per-band power and on log-variance features.
2. **Filter-bank covariance.** Per-band shrunk channel covariance matrices, mapped to the tangent space at the per-band training Fréchet mean via the matrix logarithm and vectorized for a Euclidean classifier [1].
3. **Riemannian.** Explicit affine-invariant SPD geometry (`utils/riemann.py`; geometric/Karcher mean, tangent projection, AIRM distance, trace-proportional SPD regularization, validated to Fréchet-residual $\sim 1 \times 10^{-9}$ and affine-invariance to $\sim 1 \times 10^{-14}$): a tangent-space classifier and a minimum-distance-to-Riemannian-mean (MDM) classifier, each with an optional unsupervised per-subject Riemannian recentering for label-free domain alignment [10] applied transductively to the held-out subject's own (unlabeled) inputs.
4. **EEGNet-class compact CNN** [8], implemented dependency-free in NumPy (`utils/eegnet.py`; temporal convolution $\rightarrow$ depthwise spatial convolution $\rightarrow$ separable convolution $\rightarrow$ dense classifier; $F_1 = 8$, $D = 2$, $F_2 = 16$), learning spatiotemporal filters from the raw maintenance-window waveform. The implementation passes a full finite-difference gradient check (maximum relative error $\sim 5 \times 10^{-6}$); inner per-subject early stopping is used for model selection within the training fold.

A fifth rung (EEG foundation-model transfer) was named in the plan as optional and was *not* pursued (Section 4.1).

3.3 Evaluation design

Splits. Leave-one-subject-out (LOSO) is the headline. All model selection—channels, bands, windows, thresholds, hyperparameters—happens inside the training subjects only; the held-out subject is touched once, for scoring. Subject-based cross-validation is treated as a hard requirement for a cross-subject claim, and non-nested tuning is avoided, in line with current guidance on EEG partitioning [5] and on cross-validation caveats in small neuroimaging datasets [9]. Within-subject decoding is reported only as a diagnostic ceiling, never as the headline.

Controls (all pre-declared). Label-shuffle (null for the decoding metric); *behavioral-only* (non-signal task covariates with no scalp signal and *excluding the target itself*—response time, correctness, match/mismatch, session, trial order, timing); a timing-only control; a subject-identity leakage check; and an artifact-sanity check that eye/muscle/reference channels must not dominate. The headline metric is improvement over the *strongest* non-signal control, not raw accuracy above chance.

Metric and pre-declared success bar. The primary metric is LOSO *balanced accuracy* on binary high-vs-low load during maintenance, reported per subject, with subject-level sign-flip/permutation intervals (a window-level permutation may not substitute for subject-level evidence). The success bar, fixed before any model was run, is:

(i) mean LOSO balanced-accuracy improvement $\geq +0.075$ (absolute) over the strongest non-signal control;
(ii) at least 7 of 9 held-out subjects above that control; *and*
(iii) no single-subject removal drops the mean improvement below +0.04 (a "not carried by one subject" robustness clause).

The mechanism half additionally requires ≥ 5 subjects with adequate MTL coverage, determined by a coverage audit *before* mechanism analysis.

Pre-declared outcome shapes. A "clean failure" (within-subject decoding may exist but LOSO does not beat the controls) and a "decoding-without-validated-mechanism / inconclusive" shape were both written down as allowed outcomes before any result was observed. The two-part result reported here is the resolution of those pre-declared shapes, not a post-hoc reframing (Section 4).

3.4 Mechanism layer

A coverage audit (`scripts/audit_mtl_coverage.py`) counts subjects with adequate MTL coverage from the intracranial electrode anatomy. For qualifying subjects, the mechanism probe (`scripts/run_mtl_bandpower_probe.py`, sharing a single hippocampus/amygdala/parahippocampal anatomy definition through `utils/mechanism.py`) loads MTL contacts, computes maintenance-window theta and alpha log-power, summarizes subject-level high-minus-low load effects, and correlates a supplied scalp-decoder score against the MTL bands. A residual-coupling probe (`scripts/run_mtl_residual`) then re-estimates the coupling after regressing out, in stages, the load label, the task-schedule structure (previous-trial correctness, trial index, session), and the behavioral covariates (correctness/match/RT). This residualization is the guard that prevents a raw load-linked or schedule-linked correlate from being reported as a deep readout.

4 Results

4.1 Part A — Decoding: a clean negative across the full model ladder

Table 1 reports the mean LOSO balanced accuracy of every rung on the locked 8-channel montage. No rung clears the strongest non-signal control (behavioral-only, 0.593) by the pre-declared margin; only EEGNet exceeds it on the mean at all.

Table 1: Decoding ladder, mean LOSO balanced accuracy (chance = 0.50; locked 8-channel montage). Strongest non-signal control (behavioral-only) = 0.593. "rc" = unsupervised Riemannian recentering.

Rung / model	Mean LOSO balanced accuracy
Rung 1 — logistic (per-band power)	0.560
Rung 1 — covariance (tangent features)	0.559
Rung 2 — filter-bank tangent space	0.558
Rung 2 — tangent + rc	0.552
Rung 3 — MDM	0.533
Rung 3 — MDM + rc	0.545
Rung 4 — EEGNet	**0.616**
Strongest non-signal control (behavioral-only)	0.593

The strongest control is a real, pre-declared task-schedule channel. A behavioral-control ablation (Table 2) shows the behavioral-only control is almost entirely *previous-trial correctness*. Response time, correctness/match, trial index, and session each decode at chance; previous-trial correctness alone reaches 0.596 (9/9 subjects above chance). The mechanism is the task rule: when the previous trial was incorrect, the current trial is high-load only 2/130 times (0.015); when it was correct, the current trial is high-load 1021/1523 times (0.670). This is not response-time leakage, correctness-as-performance leakage, or session/trial drift; it is a genuine task-schedule control channel that the protocol pre-declared as an allowed control, so it stands as the strongest non-signal baseline that a scalp decoder must beat.

Table 2: Behavioral-control ablation, mean LOSO balanced accuracy. The behavioral-only control is carried by previous-trial correctness, a task-schedule channel.

Component	Mean LOSO balanced accuracy	Subjects > 0.50
Response time only	0.500	0/9
Correctness + match/mismatch	0.500	0/9
Previous-trial correctness only	0.596	9/9
Trial index only	0.500	0/9
Session only	0.500	0/9
Trial index + session	0.500	0/9
Full behavioral control	0.593	9/9

EEGNet beats the control on the mean but fails the robustness clause. Table 3 details the headline EEGNet rung against the success bar. EEGNet is the only rung to exceed the behavioral control on the mean (0.616 vs 0.593, all nine subjects above chance), so the prior expectation that no model would beat the control was wrong on the mean. But it fails the bar, and fails it in exactly the way the robustness clause was written to catch: the positive mean rests entirely on a single subject (S04, improvement +0.218; next-best +0.045). Removing S04 collapses the leave-one-subject-removed mean improvement to −0.001. Only 5/9 subjects are above their strongest control, the bootstrap 95% confidence interval for the mean improvement crosses zero, and the subject-level sign-flip test is not significant.

Table 3: Headline EEGNet rung against the pre-declared success bar. Every criterion fails.

Quantity	Value	Bar	Pass?
Mean LOSO balanced accuracy (signal)	0.616	—	—
Mean strongest-control balanced accuracy	0.593	—	—
Mean improvement	+0.023	$\geq +0.075$	No
Subjects above strongest control	5/9	$\geq 7/9$	No
Min leave-one-subject-removed mean improvement	−0.001	$\geq +0.04$	No
Bootstrap 95% CI for mean improvement	$[-0.022, +0.081]$	excludes 0	No (crosses 0)
Subject sign-flip p (one-sided)	0.262	—	n.s.
Headline success	—	—	**No**

The constraint is the montage and cross-subject transfer, not model capacity. Two diagnostics localize the wall. First, the subject-identity diagnostic is 1.000 on every fold: the covariance feature space is perfectly subject-separable, the regime in which added model capacity buys subject *identity* rather than transferable load signal. Second, the pre-declared reference-artifact

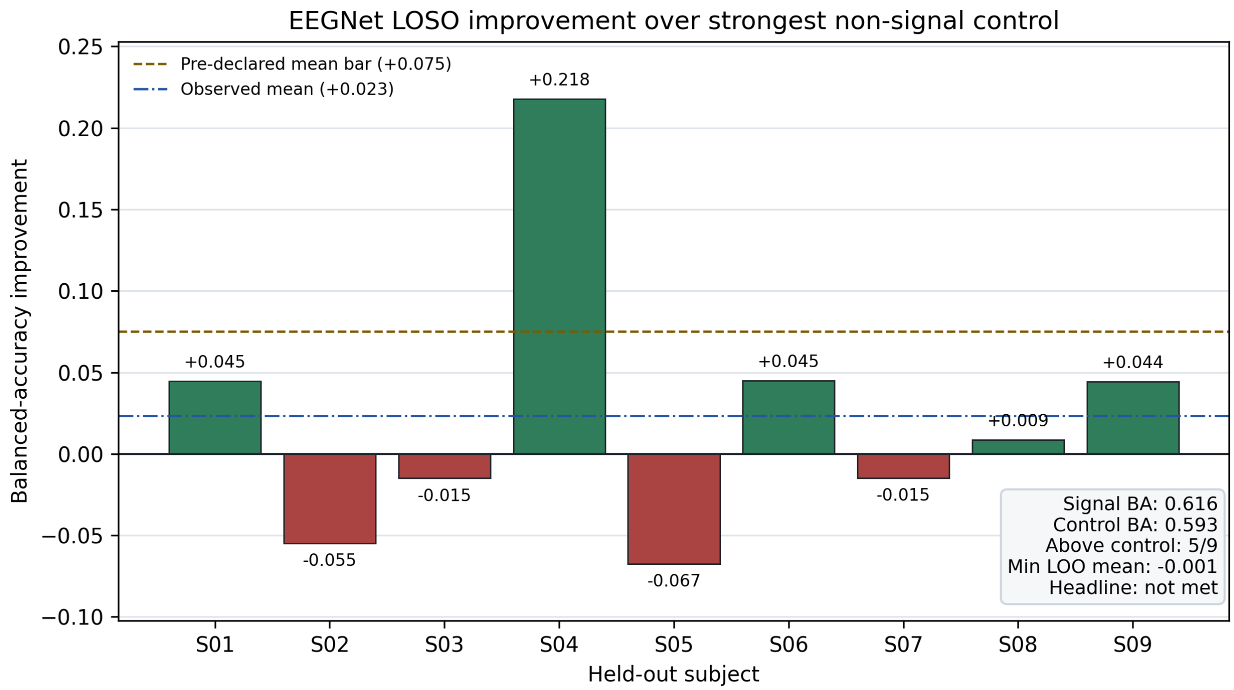

Figure 1: Dashboard-derived per-subject EEGNet improvement over the strongest non-signal control. The observed mean improvement is positive but below the pre-declared bar, and the positive mean is carried by S04.

check passes: brain-only 6-channel EEGNet reaches 0.623, essentially equal to the 8-channel 0.616, so the (sub-threshold) signal is not an artifact of the A1/A2 ear references. Because the entire pre-registered model ladder—linear, covariance, Riemannian, and a compact CNN—was thrown at the problem without clearing the bar, the optional foundation-model rung was not run: it would change the cost and the claim without addressing the binding constraint. Part A is therefore the pre-declared *clean failure*, established across the whole model class rather than a single model.

4.2 Part B — Mechanism: a real MTL substrate and an exploratory, non-surviving coupling

The MTL coverage audit qualifies all 9/9 subjects (each has hippocampal contacts; 6–21 MTL contacts each; the gate needs ≥ 5), so the mechanism layer is fully powered with respect to coverage.

The intracranial substrate is real. On the recorded MTL contacts, the theta-minus-alpha load effect is positive in 7/9 subjects with a two-sided sign-flip $p = 0.016$ (mean $z = 0.143$); the individual theta and alpha load effects are weaker and not individually significant (Table 4). This reproduces, in this dataset, the load-dependent MTL theta–alpha pattern that the coupling literature reports [2].

The strongest decoder couples to it—raw. Correlating each decoder's score against the MTL bands (Table 5), the linear/tangent decoders show essentially zero-to-negative coupling, whereas the EEGNet score—the one that extracts marginally more load information—flips to positive across theta, alpha, and the theta-minus-alpha differential, reaching 7/9 subjects at a borderline $p = 0.0508$

Table 4: MTL high-minus-low load effects (intracranial), subject-level.

Metric	Mean (z)	Positive subjects	Two-sided sign-flip p
MTL theta load effect	0.120	5/9	0.324
MTL alpha load effect	0.025	5/9	0.809
MTL theta−alpha load effect	**0.143**	**7/9**	**0.016**

on the differential. The direction is consistent: a better scalp decoder tracks recorded MTL theta dynamics more. This is the first positive evidence the project produced that the scalp load signature relates to deep MTL activity.

Table 5: Coupling between scalp-decoder score and MTL bands. The stronger (EEGNet) decoder couples where the linear/tangent decoder does not.

Coupling metric	Tangent decoder	EEGNet decoder
corr(score, MTL theta)	−0.011 (5/9, p=0.87)	+0.078 (6/9, p=0.16)
corr(score, MTL alpha)	−0.018 (3/9, p=0.82)	+0.057 (6/9, p=0.29)
corr(score, MTL theta−alpha)	−0.015 (5/9, p=0.81)	+0.068 (7/9, p=0.0508)

The coupling does not survive residualization. The raw EEGNet coupling (+0.068, 7/9, $p = 0.0508$) weakens monotonically as task structure is partialled out (Table 6; Figure 2): to +0.050 after residualizing on load, to +0.011 after additionally residualizing on the task schedule, and to +0.013 after residualizing on the behavioral covariates. After removing label and schedule structure the coupling effectively disappears. This is consistent with *either* a real load-linked shared MTL state—in which case residualizing load legitimately removes part of the genuine shared variance—*or* a schedule-linked correlate, and at $n = 9$ this dataset cannot disambiguate the two. We therefore report Part B as an **exploratory/inconclusive** lead, explicitly not a validated deep readout.

Table 6: Residual-coupling sensitivity: EEGNet score vs. MTL theta−alpha differential. Raw coupling does not survive load/schedule residualization.

Residualization	Mean corr	Positive subjects	Two-sided sign-flip p
Raw	+0.068	7/9	0.0508
On load	+0.050	5/9	0.133
On load + schedule (prev-correct, trial idx, session)	+0.011	4/9	0.746
On load + schedule + behavior (correctness/match/RT)	+0.013	5/9	0.715

Confirmatory coupling gate. Codex operationalized the prospective Part B gate in `scripts/run_mtl_confirma` The gate fixed the metric to the schedule-residualized EEGNet score versus MTL theta-minus-alpha differential and required all of the following: positive mean correlation, at least 7/9 positive subjects, exact two-sided subject sign-flip $p \leq 0.05$, and every leave-one-subject-out mean above zero. It failed clearly: mean +0.011, 4/9 positive subjects, $p = 0.7461$, and minimum leave-one-subject-out mean −0.010. The confirmatory result therefore does not rescue the mechanism half. Part B remains an exploratory mechanism lead, not validated deep-source readout.

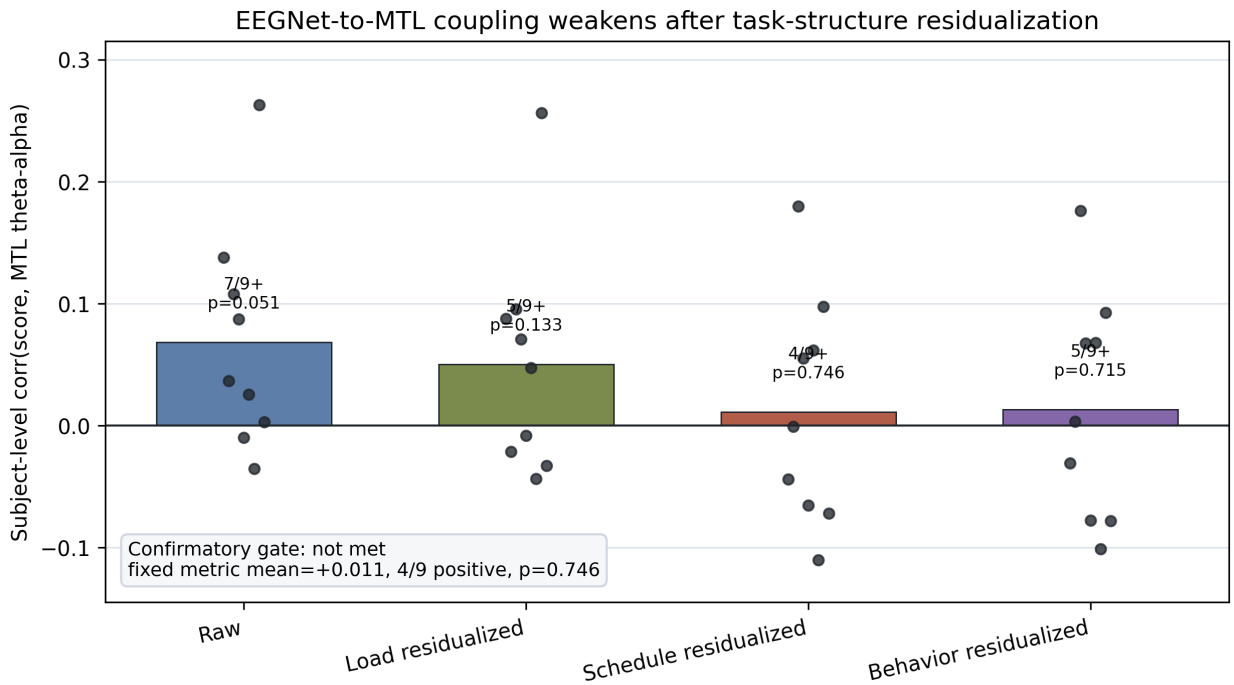

Figure 2: Dashboard-derived mechanism sensitivity figure. The raw EEGNet score-to-MTL theta-minus-alpha coupling is suggestive, but the fixed schedule-residualized metric used for the confirmatory gate is near zero and fails the subject-count, sign-flip, and leave-one-subject-out robustness criteria.

5 Limitations

- **Nine subjects.** The dominant statistical constraint. Subject-level tests at $n = 9$ have wide error bars; a 7/9, $p \approx 0.05$ effect is suggestive, not conclusive, and small decoder gains can be carried by a single subject—exactly what the robustness clause caught for EEGNet [9].
- **Common montage.** The 8-channel common montage is what makes a single cross-subject model possible, but it is also a likely cause of the decoding wall. A richer per-subject channel set, or within-subject calibration, would be a different claim (within-subject, not transferable), not a rescue of this one.
- **Maintenance window and load contrast.** A single pre-declared maintenance window and a binary 4-vs-6/8 load contrast were used for the headline; other windows and graded set-size targets are diagnostics only.
- **Mechanism direction.** The residualization ambiguity (real shared load state vs. schedule-linked correlate) is intrinsic to this dataset's size and task schedule; resolving it needs more subjects or a task without the incorrect-then-low-load rule.
- **Clinical scope.** De-identified public research data; no human-subjects action and no medical or diagnostic claims. This is a measurement study.

Quality control and exclusions. Per the project standards, exclusions are named rather than silent. No subject is excluded from the headline decoding (all nine enter LOSO); all nine also qualify for the mechanism layer by the coverage audit. The brain-only 6-channel run is a deliberate sensitivity variant, not an exclusion. Retained-trial counts, per-subject scores, and all control outputs are preserved in the Reproducibility Packet so any exclusion or filter is auditable.

6 Conclusion

We set out to test the smallest honest first rung toward electrical fMRI: whether scalp EEG carries a subject-transferable, intracranially-validated signature of MTL working-memory load. The answer, reported as two pre-declared parts, is a bounded negative plus an exploratory lead. **Part A:** an 8-channel common montage does not support transferable load decoding above the strongest non-signal control across a complete model ladder—linear, covariance, Riemannian, and a compact CNN—and the binding constraint is the montage and cross-subject transfer, not model capacity. **Part B:** the MTL theta-minus-alpha load substrate is real and the strongest scalp decoder couples to it in raw trial summaries where weaker decoders do not, but that coupling does not survive load/schedule residualization at $n = 9$ and is reported as exploratory, not validated.

For the larger program this is a useful, honest result. It locates the first wall—8-channel common-montage cross-subject load decoding—and it names the most plausible next signal to pursue: the MTL theta–alpha coupling that the strongest decoder began to track, tested with more statistical power, a richer or subject-specific montage, or a task design that breaks the correctness-to-load schedule confound. Both outcomes were named before any result was seen; reporting which pre-declared shapes resolved, rather than reframing after the fact, is the discipline that makes the negative a contribution rather than a disappointment.

Data and code availability

The dataset is the public Boran et al. (2019) release [3] (G-Node DOI `10.12751/g-node.d76994`, CC BY-SA 4.0); raw data is not redistributed here. All analysis code, pinned dependencies, and the per-subject verification dashboard ship in the project's Reproducibility Packet, which reproduces every number in this report end-to-end from the downloaded dataset.

Acknowledgments

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